FAIRSpectra

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SurfaceSpectra

A Better File Format for SIMS Data

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FAIRSpectra Initiative

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https://alexhenderson.info

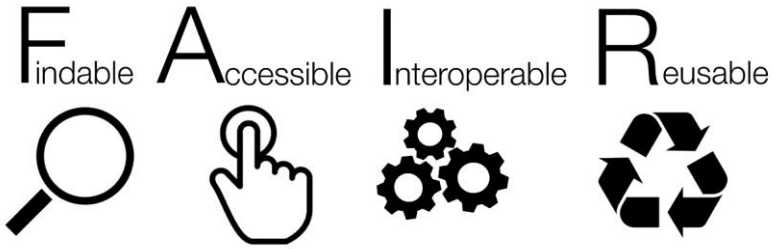
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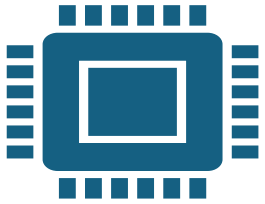
What is FAIR?

The FAIR Guiding Principles

Findable
Accessible
Interoperable
Reusable



https://www.go-fair.org



Interoperable

Integration with other data, applications and workflows for analysis and storage



Reusable

Well-described so they can be replicated and/or combined in different settings

FAIRSpectra Initiative

Community-driven initiative

Focus on hyperspectral imaging techniques

- File formats for hyperspectral imaging
  - No standards exist right now
  - Software tools to support these
- Metadata requirements
- Education and training
- Raising awareness



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What do we need?



Perfect data format?

- Efficiently packaged binary data
- Fast to write and read
- Cloud-friendly
- Sufficient metadata to be understood
- Provenance recorded
- AI-ready
- FAIR

Photo by Ben White on Unsplash



What do we want a format to enable?

Depends on the context

- Visualization
- Multivariate statistics / ML / AI
- Create databases
- Library search
- Data fusion with other techniques
- Use existing software from elsewhere
- Other, yet to be determined…

What do these contexts require?

Target	Requires
Visualization	Quick access to data. 1D, 2D, 3D
Multivariate statistics / ML / AI	All data in RAM, or out-of-core?
Create databases	Access metadata separate from spectra
Library search	Accurate mass
Data fusion with other techniques	Coordinate systems to match
Use existing software from elsewhere	Open platform

Metadata

Information about the data

Which metadata are required?

Upstream sample provenance

Sample

- Sampling method
- Storage conditions
- Chemical modifications
- Physical state
- Pre-treatment
- …



Experiment

- Experiment plan
- Substrate material
- Mounting method
- Region analysed
- Instrument params
- …



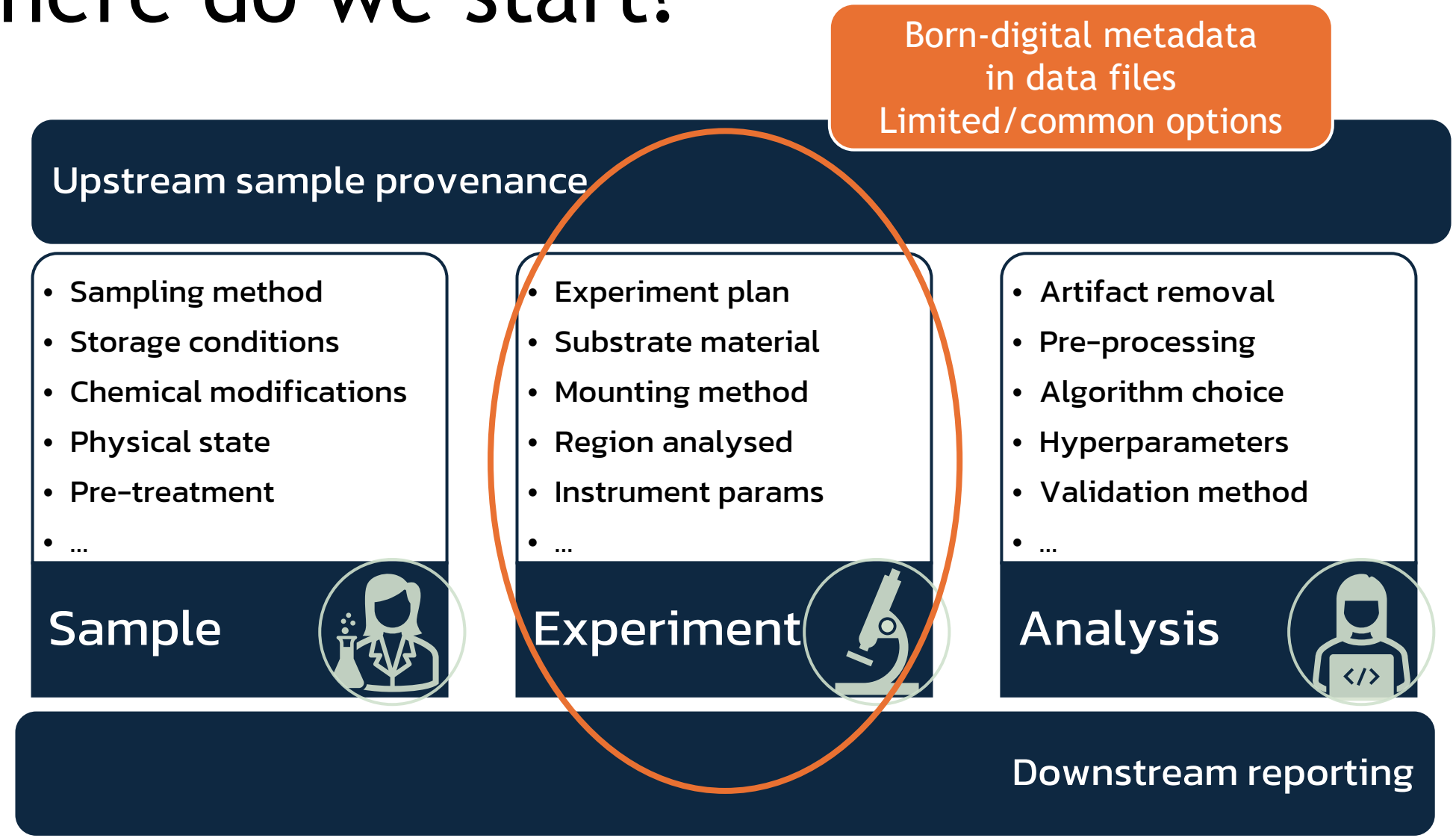
Analysis

- Artifact removal
- Pre-processing
- Algorithm choice
- Hyperparameters
- Validation method
- …

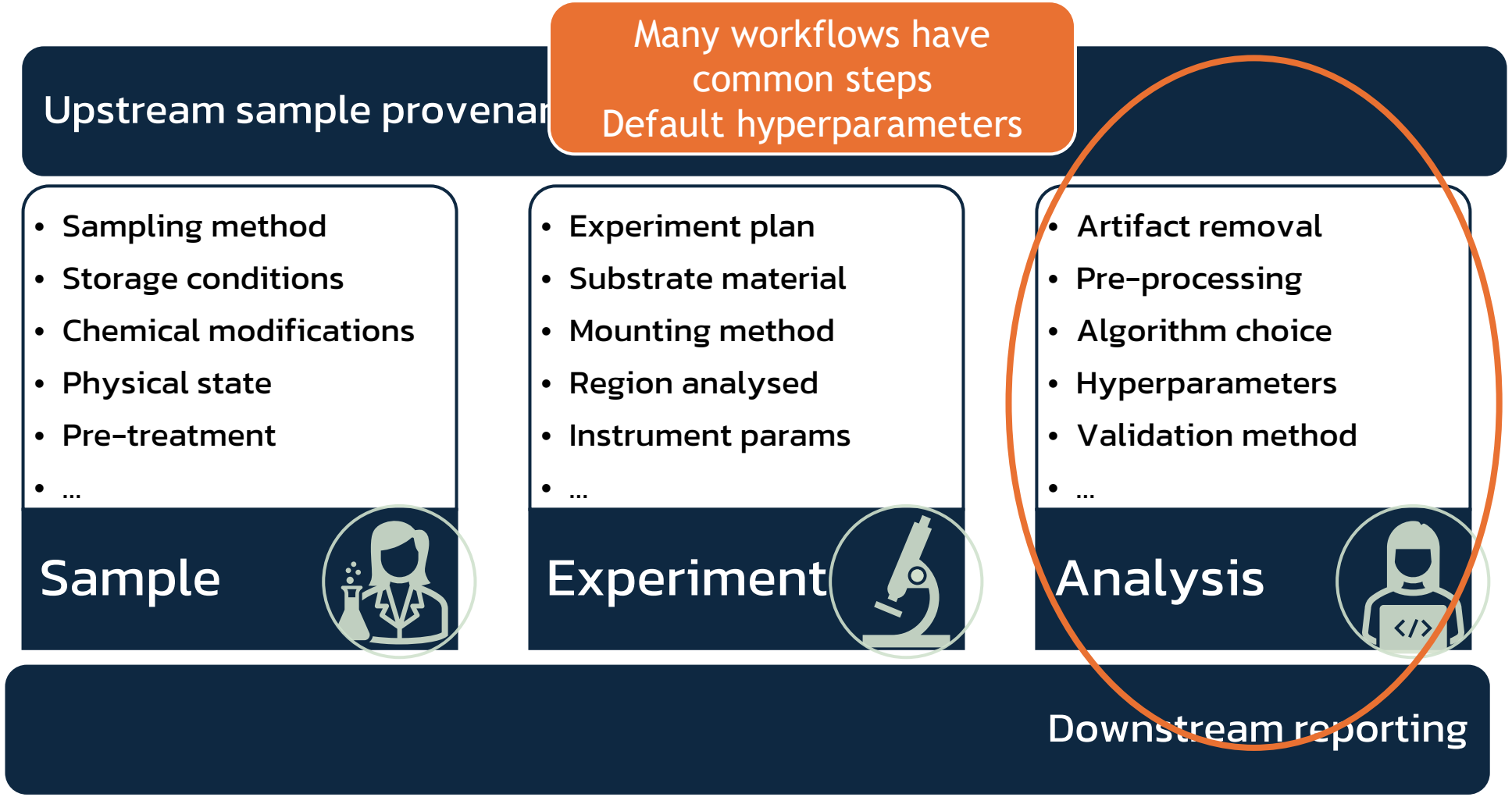


Downstream reporting

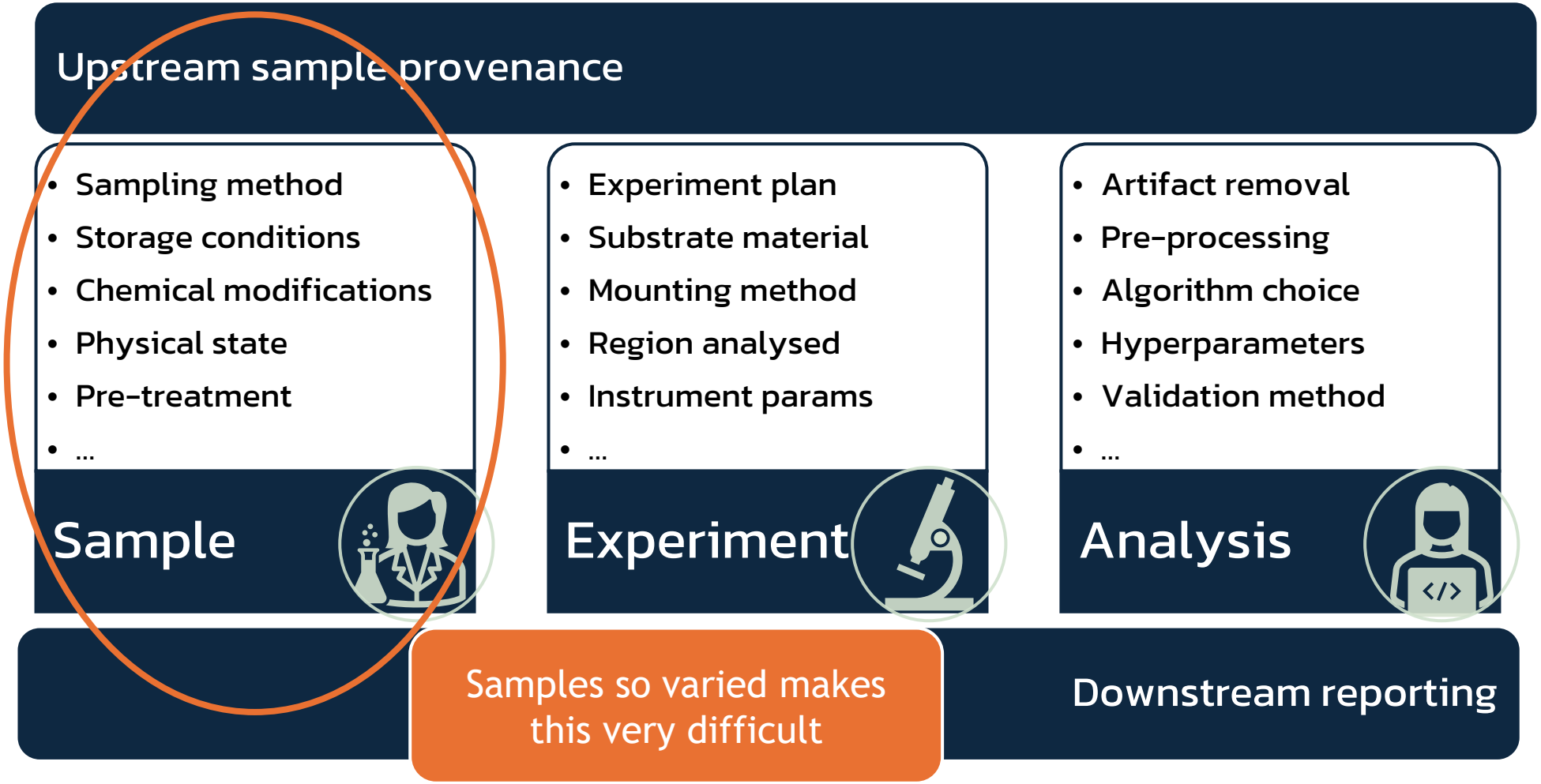
Where do we start?



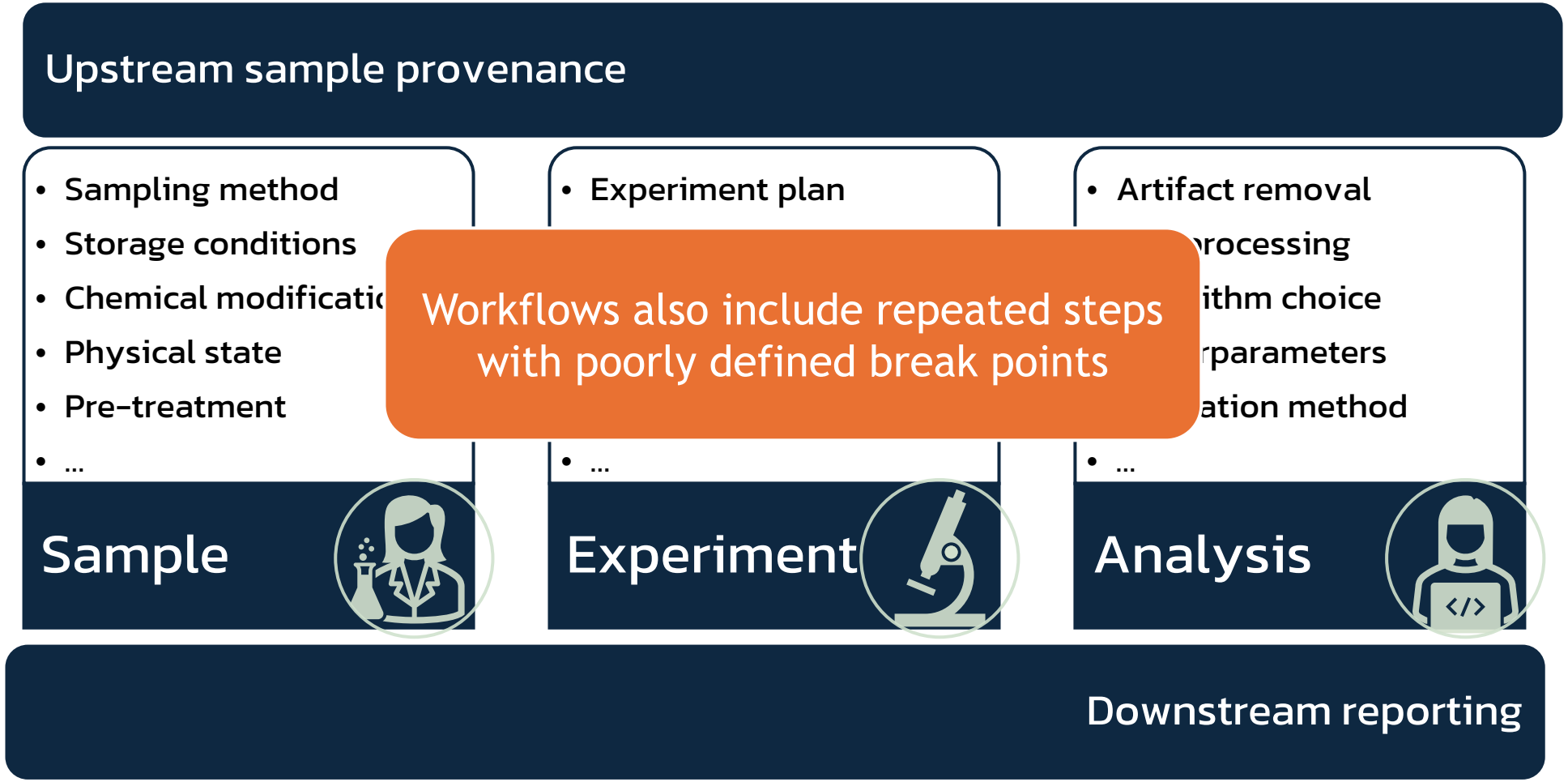
Where do we start?



Where do we start?



Where do we start?

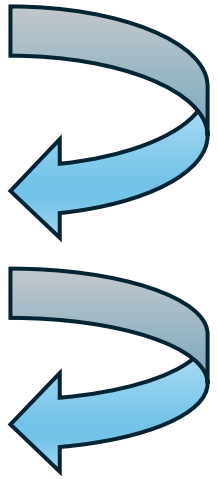


Lessons from the web community

Unambiguous terms (good **semantics**)

Resource Description Framework (**RDF**)

- Based around the **graph representation** of information



Combine RDF-encoded metadata into **Linked Data**

- Enables **interconnectivity** between samples/instruments/SOPs

Good for unstructured data

Promotes FAIR practice

Semantics is the study of ‘meaning’

Standard terminologies for surface analysis

ISO:18115

- Part 1: General terms and terms used in spectroscopy
- Part 2: Terms used in scanning-probe microscopy
- Part 3: Terms used in optical interface analysis

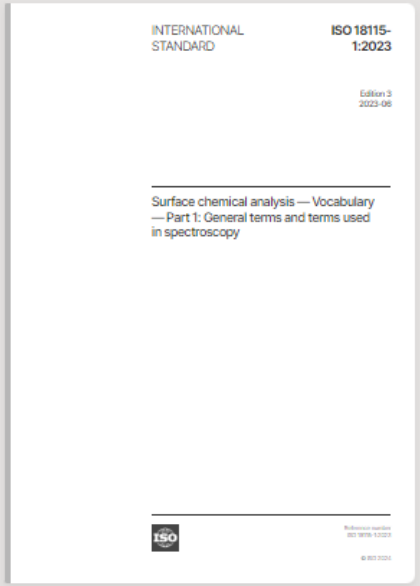
Other documents are in development. Please get involved!

https://www.iso.org/standard/74811.html

Applications OBP English

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11-22 Nov 2024: Follow ISO at COP29



Read sample

ISO 18115-1:2023

Surface chemical analysis — Vocabulary

Part 1: General terms and terms used in spectroscopy

Published (Edition 3, 2023)

ISO 18115-1:2023

CHF 42

Language

English

Format

- PDF + ePub
- Paper

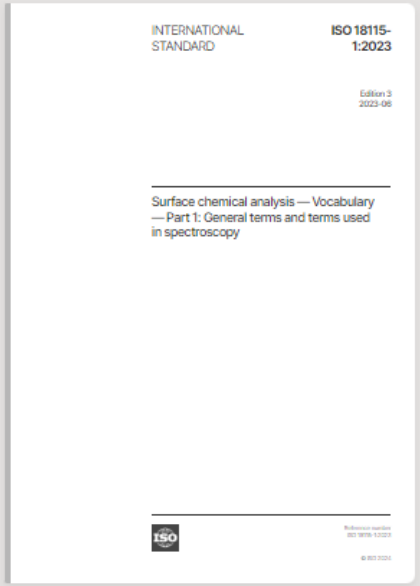
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https://www.iso.org/standard/74811.html

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Read sample

ISO 18115-1:2023

Surfa
— Vo

Part
used

FREE!

Published (Edition 3, 2023)

ISO 18115-1:2023

42

Language

English

Format

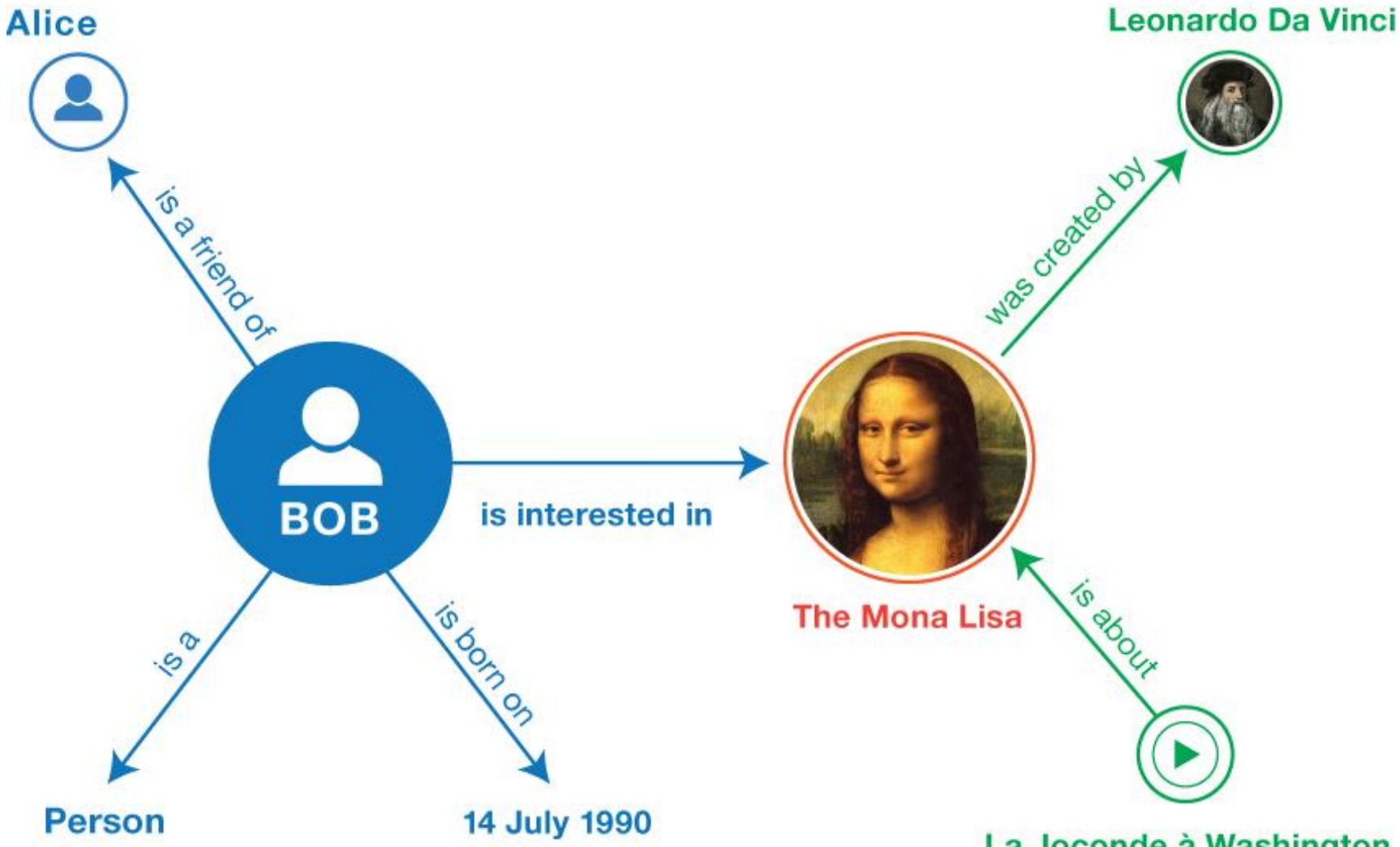
PDF + e

Paper

Add to cart

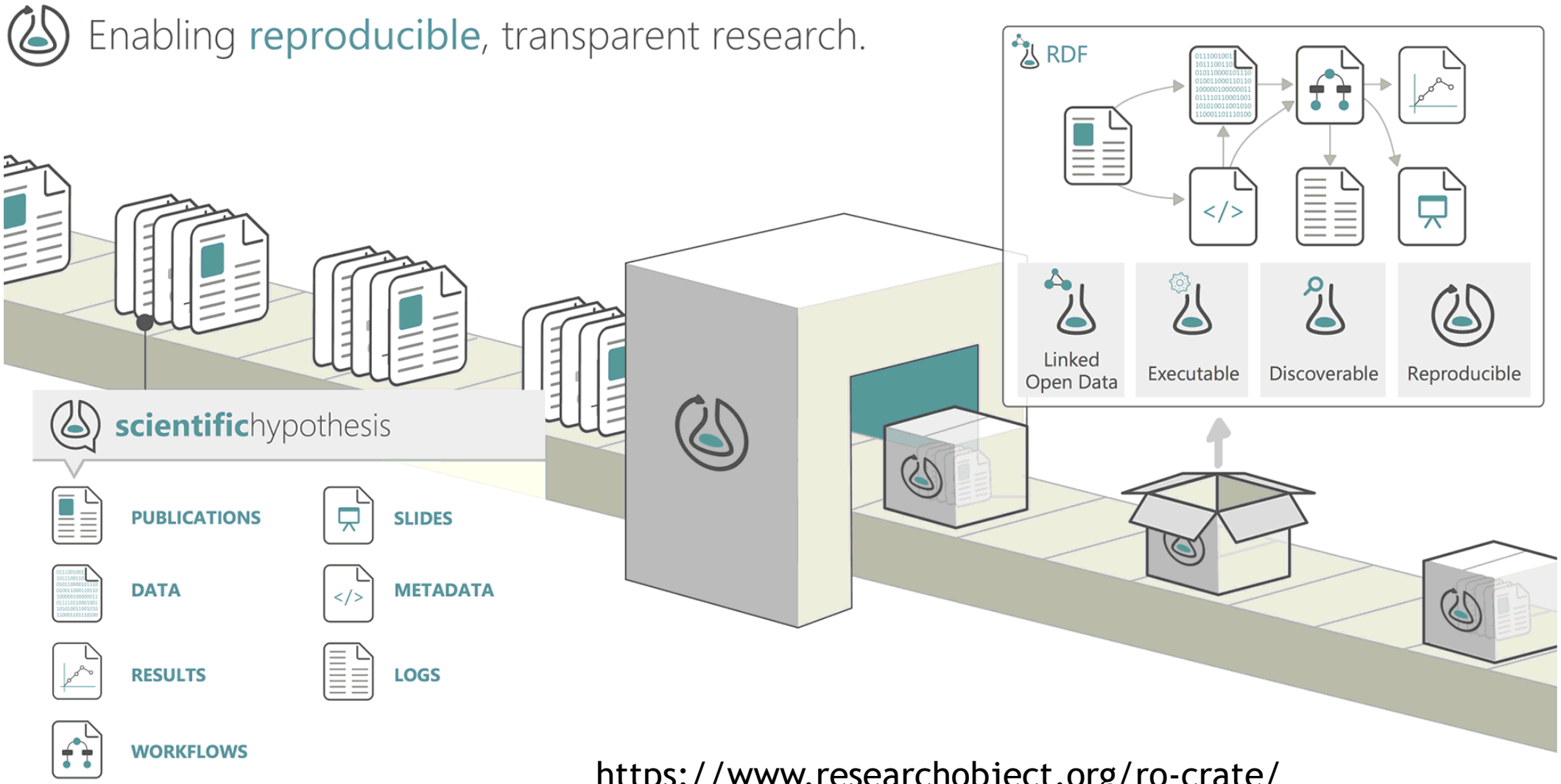
Linked Data

Data can be validated against minimum reporting requirements (SHACL, ShEx, Link-ML)



https://w3c.github.io/rdf-primer/spec/

Research Object Crate (RO-Crate)



https://www.researchobject.org/ro-crate/

RO-Crate

- Implementation of a FAIR Digital Object (FDO)
- Based on **schema.org** developed by Google, Microsoft and others for the semantic web
- Weak semantics — can use free text if semantic term not available
- BYO — Bring Your Own Ontology

Adaptable | Flexible | Expandable

Metadata LEGO



Develop LEGO™-style bricks of metadata info

Make these semantic (understandable by machines)

Assign unique identifiers with central registry

Aggregate bricks to create standard operating procedures

- Citeable
- Machine actionable
- Version controlled

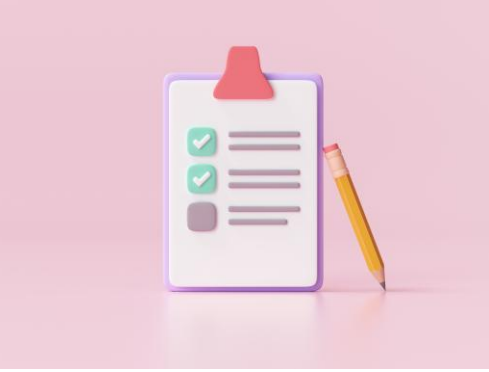
Example: Bi_3^+ primary ion


```
{"@context": "https://w3id.org/ro/crate/1.3/context"
 …
  {
    "@id": "Bi3+",
    "@type": "MolecularEntity",
    "name": "Bi3+",
    "description": "A cluster of 3 bismuth atoms, with a single positive charge.",
    "inChI": "InChI=1S/Bi3/q+1",
    "hasRepresentation": "[Bi<sub>3</sub>]<sup>+</sup>"
  }
}
```

Example: Water-6k^{+} primary ion


```
{"@context": "https://w3id.org/ro/crate/1.3/context"
 …
  {
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    "@type": "MolecularEntity",
    "name": "(H2O)6000+",
    "description": "Approximately 6000 water molecules in a giant gas cluster,
                    with a single positive charge.",
    "hasRepresentation": "[(H<sub>2</sub>O)<sub>6000</sub>]<sup>+</sup>"
  }
}
```

Provenance




```
{
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  {
   "@id": "ro-crate-metadata.json",
   "@type": "CreativeWork",
   "conformsTo": {"@id": "https://w3id.org/ro/crate/1.3"},
   "about": {"@id": "./"}
  },
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   "url": "http://manchester.ac.uk/"
  },
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   "@type": "Individual Product",
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   "description": "Ionoptika J105 3D Chemical Imager at the University of Manchester.",
   "url": "https://www.sarc.manchester.ac.uk/instrumentation/j105/",
   "serialNumber": "j007",
   "manufacturer": "https://ionoptika.com/"
  }
.
.
.
 ]
}
```


Work in progress – incomplete RO-Crate

Validation

- Linked-data is a graph
- Supports validation via RDF-SHACL

- Use RO-Crate ‘profile’ to implement minimum reporting requirement

- Validate metadata against profile using LinkML



Community Resource

- ‘Bring your own’ metadata LEGO
- Develop SOPs from own papers
- Donate/recommend metadata



Binary payload

The zeros and ones of our spectral data

The problem with SIMS

Answers on a postcard to…

fairspectra.net

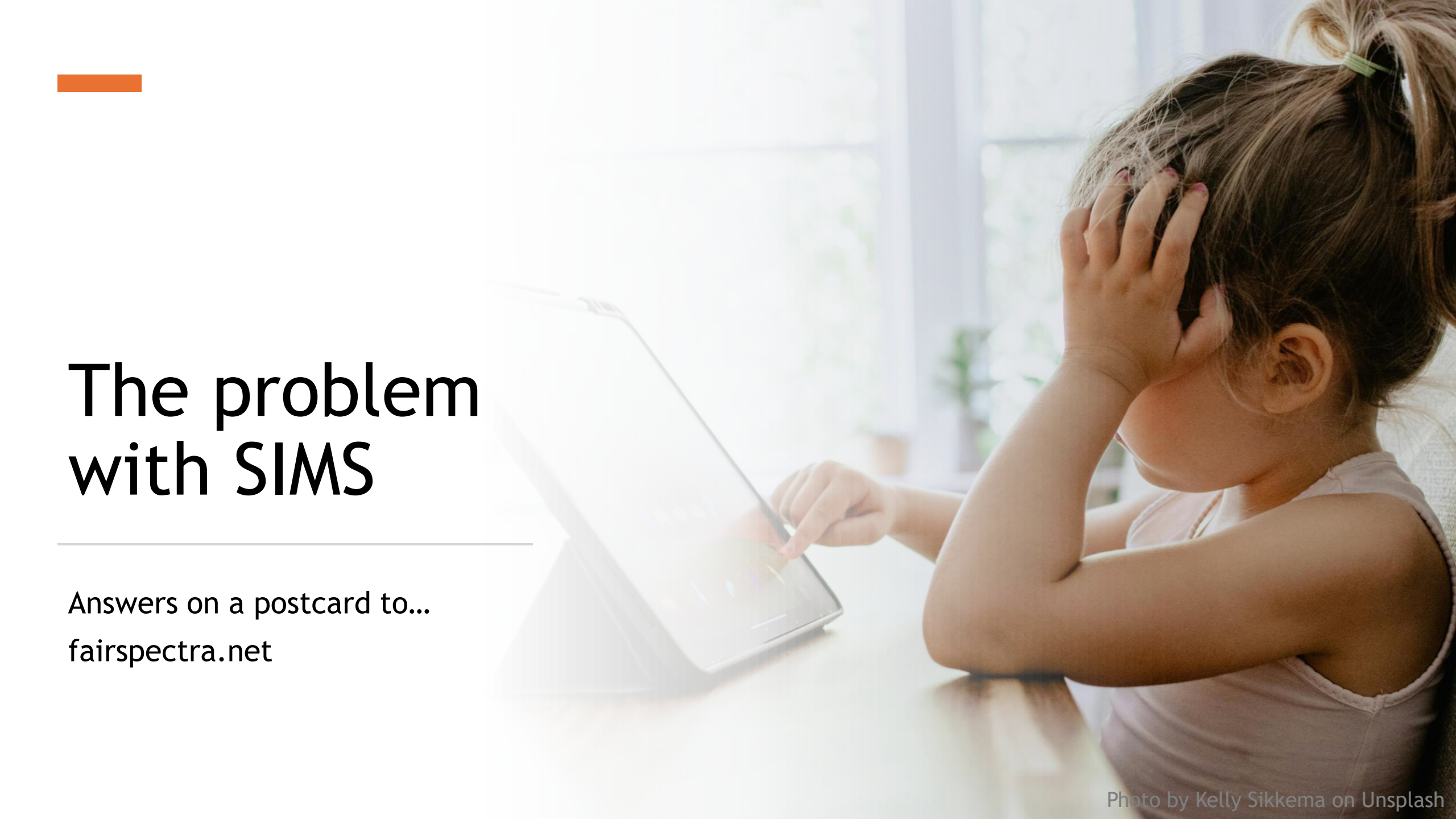


Photo by Kelly Sikkema on Unsplash

SIMS data characteristics



- Huge number of data channels
- Very sparse – almost all data channels have zero counts
- Non-zero values are ‘clumped‘ together → peaks

Potential solutions

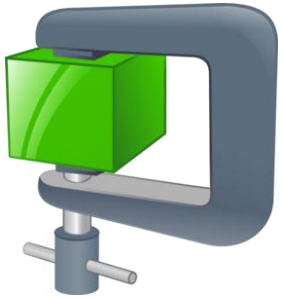


Photo by Neil Thomas on Unsplash

Solutions to the sparsity problem

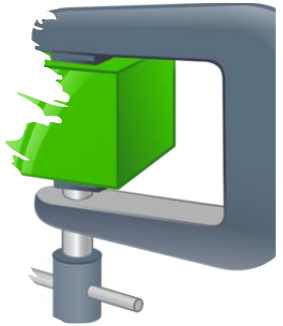
Lossless compression

- Compress and decompress returns original data unchanged
  - Use off-the-shelf lossless compression (e.g. ZIP)
  - Only record non-zero events as coordinates



Lossy compression

- Some data discarded during compression
  - Use off-the-shelf lossy compression (e.g. MP3-like, or JPEG-like)
  - Down-bin spectra to lower mass resolution
  - Use peak detection and only record centroid and area



Problems with lossless compression

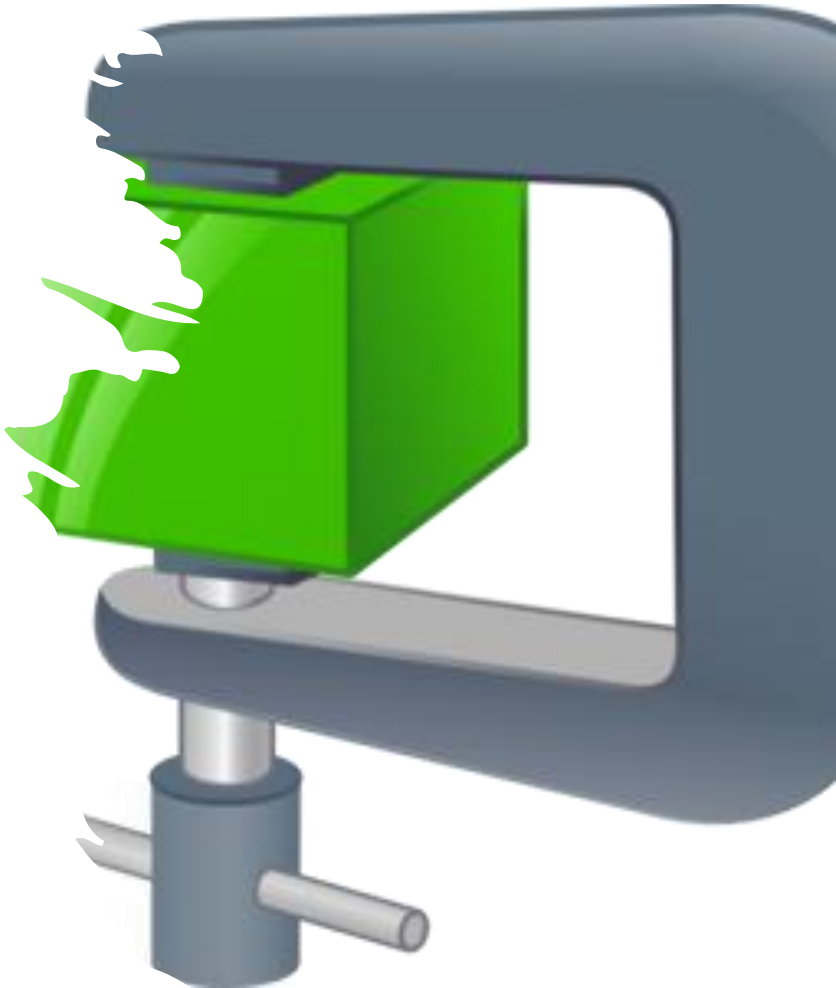
- May need to unpack/unzip data to work on it
- Compression methods not tuned to SIMS data
  - Designed for floating point numbers, or text
- Bespoke encoding requires bespoke software
  - Need to reinvent the wheel for each algorithm



Image by Freeimages.com

Problems with lossy compression

- Throwing away data
- Possible loss of mass- or spatial resolution
- Not possible to ‘round-trip’ data



imzML

Developed by MALDI community

Metadata – largely biological (HUPO)

Data stored in two flavours

- Processed – discrete peaks/regions
  - Who decided the peak positions? Parameters?
  - Data locked in
- Continuous – all spectral channels, no compression
  - Data becomes unwieldy/impossible to store

No facility for 3D data

Alternative data encoding

Formats like HDF5 and Zarr can be **chunked** and **compressed**



Photo by Markus Spiske on Unsplash

Methods from other fields

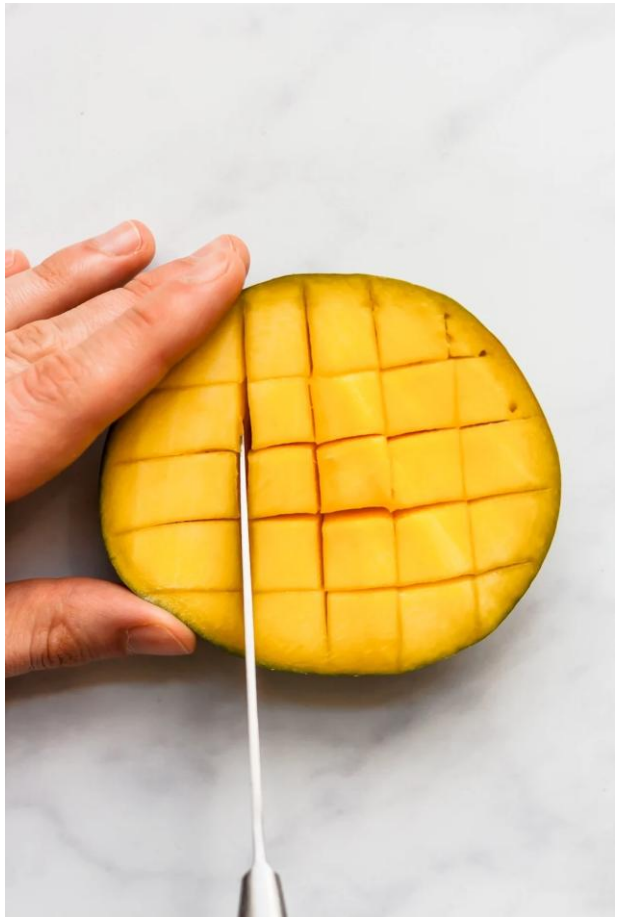
Astronomy – HDF5
Climate research – netCDF, HDF5, Zarr
Microscopy – OME-NGFF, OME-Zarr



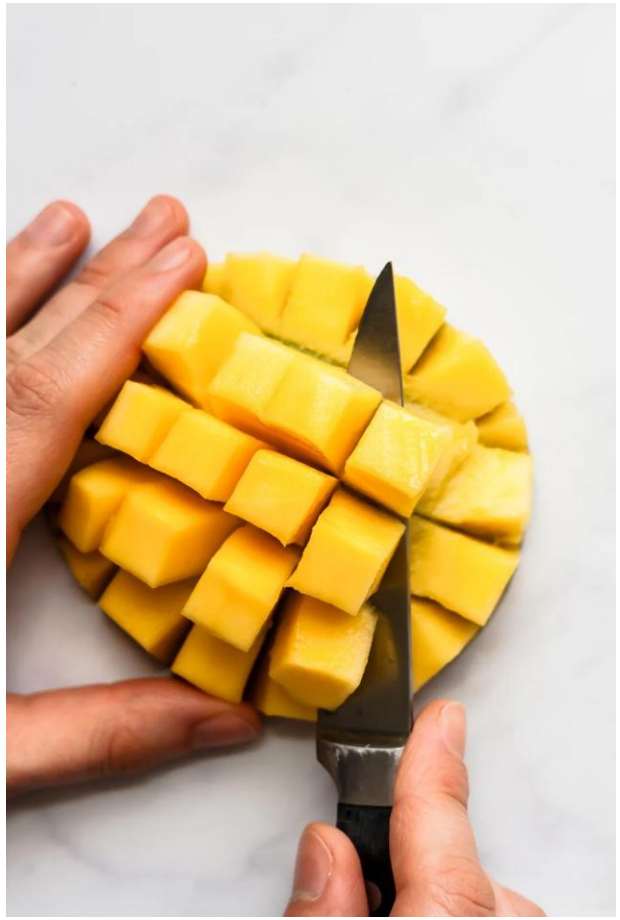


- Chunked formats
- Built-in lossless compression
- Plugin compression methods

Chunking of a hyper-mango

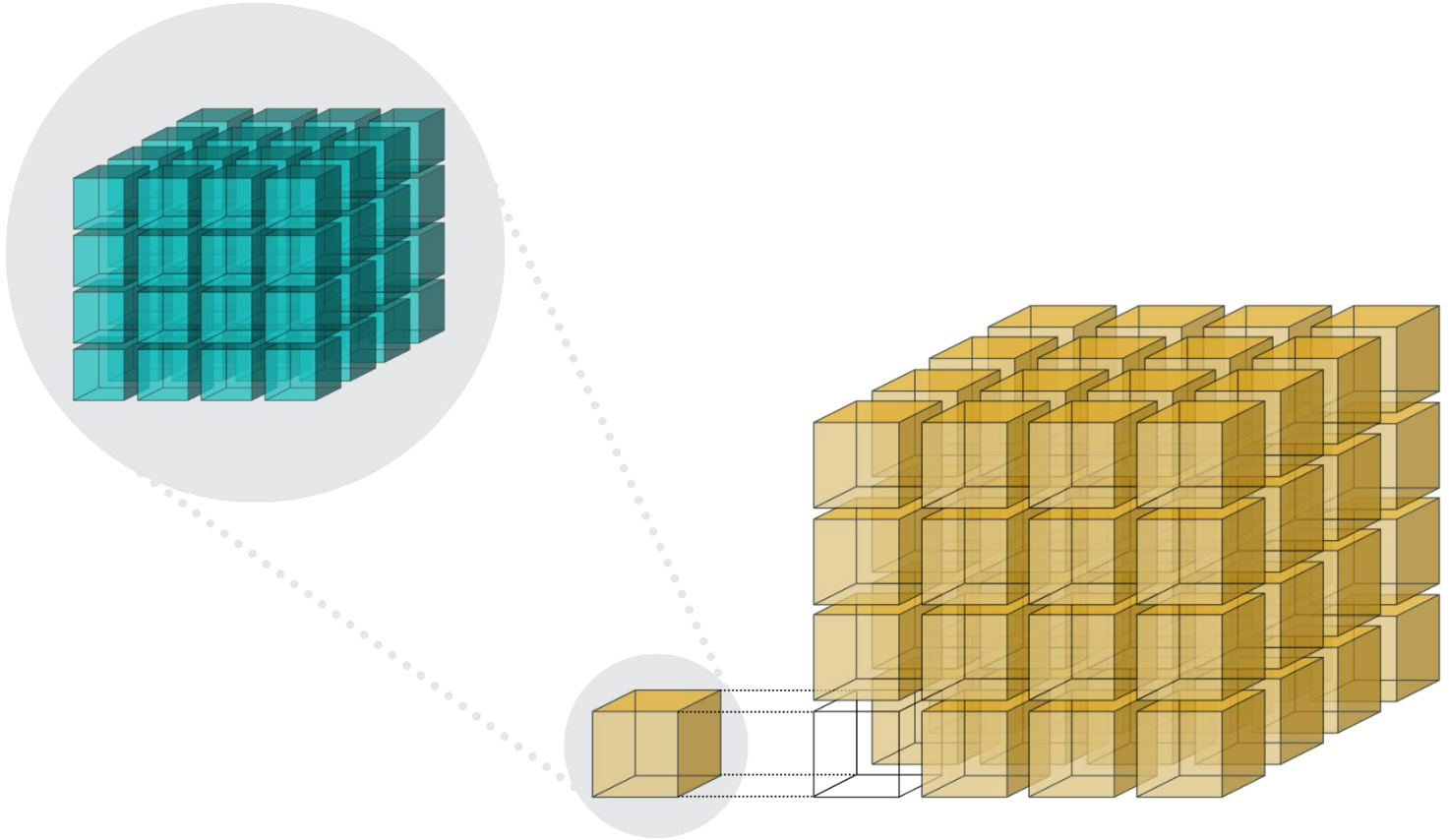






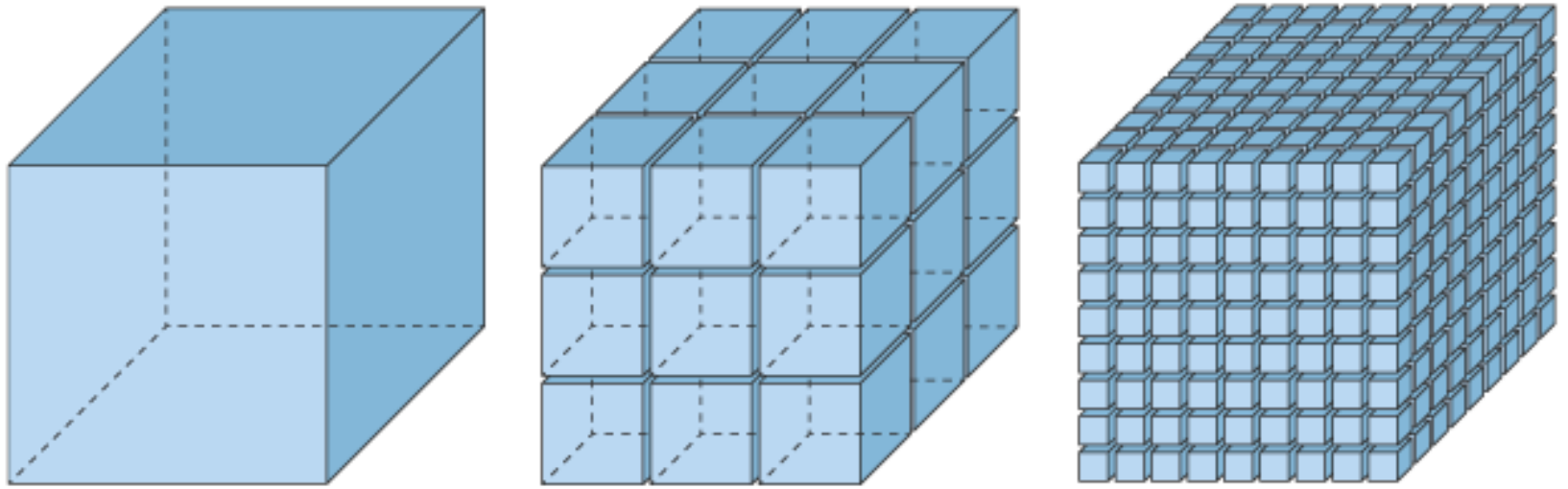
https://www.ambitiouskitchen.com/how-to-cut-a-mango/

Each mini-cube is separately addressable and compressible



https://www.blosc.org/posts/blosc2-ndim-intro/

Different chunk sizes



Smaller chunks gives higher granularity in selection, but more addressable segments increases data size

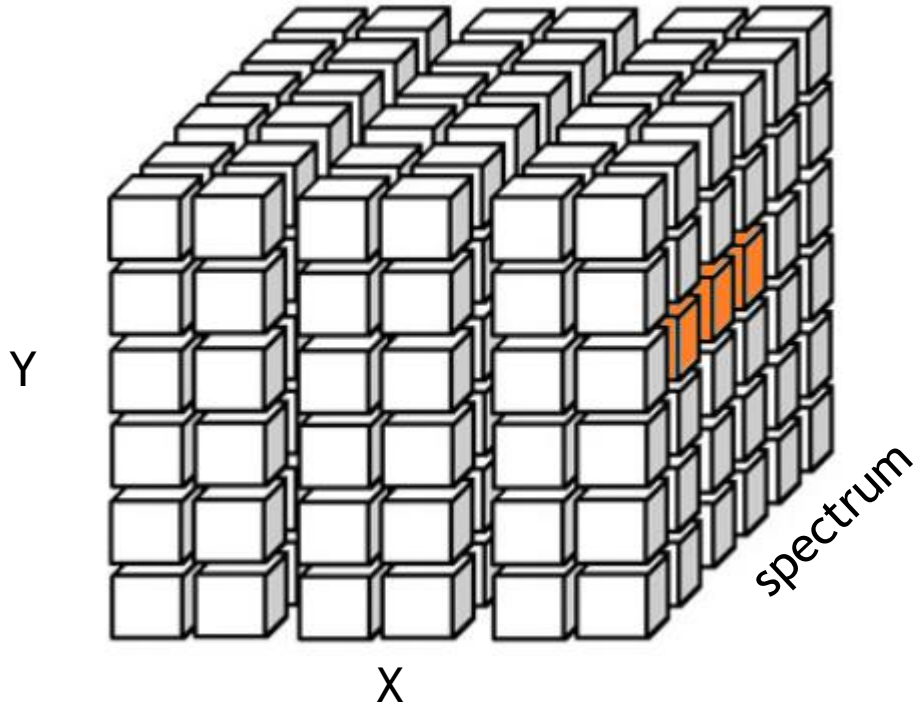
Chunks can be compressed. Contents of a chunk changes compressibility.

Difficult to predict overall file size

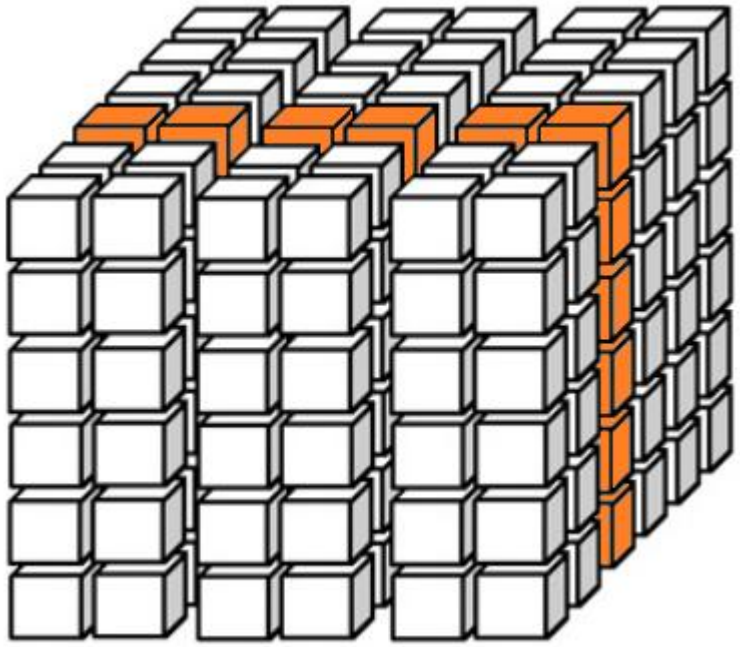
https://www.bbc.co.uk/bitesize/guides/zjs9dxs/revision/3

Only relevant segments are loaded to RAM → cloud storage friendly

Spectral range



Image

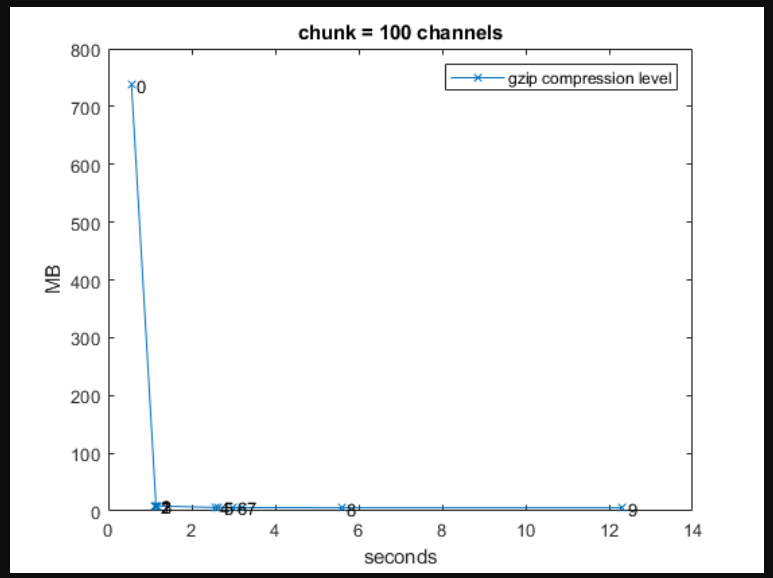


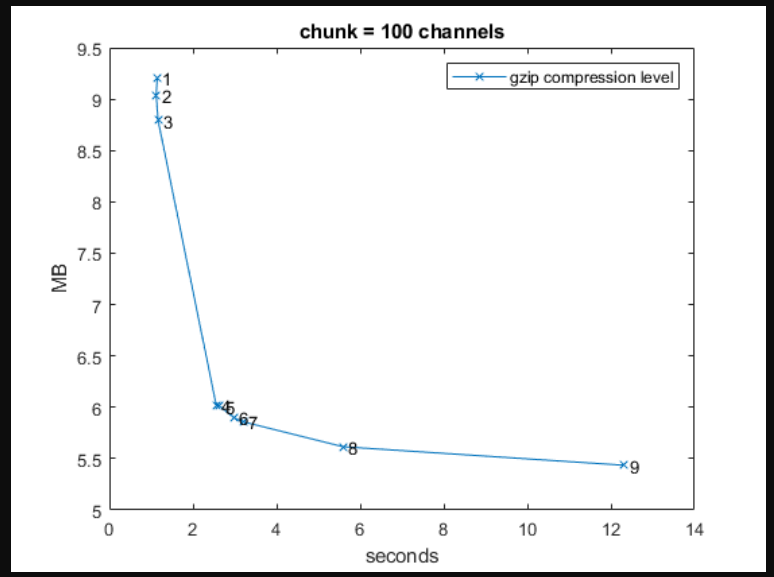
Segments are cached and garbage collected

https://commons.wikimedia.org/wiki/File:OLAPcube.png

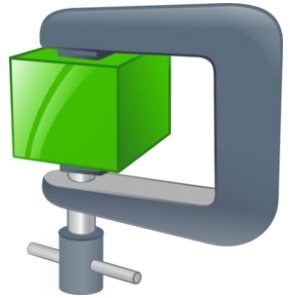
HDF5 with gzip compression

- ANY compression helps
- Data can be recompressed to balance access time and file size
- No change to data analysis approach





Sparse compression



Prototype HDF5 compression filter ready to test

Two types of compression included:

- Run-length encoding (RLE)
  - Replace runs of zero values with a single negative number
  - Guaranteed not to increase the size

0, 0, 1, 5, 2, 0, 0, 0, 0, 2, 3, … → -2, 1, 5, 2, -4, 2, 3, …

- Dictionary of keys (DOK) sparse encoding
  - Only record positions of non-zero values in pairs

0, 0, 1, 5, 2, 0, 0, 0, 0, 2, 3, …
→ {3, 1}, {4, 5}, {5, 2}, {10, 2}, {11, 3} …

Summary

Progress is slow but steady

Perfect data format?

- Efficiently packaged binary data
- Fast to write and read
- Cloud-friendly
- Sufficient metadata to be understood
- Provenance recorded
- AI-ready
- FAIR



Photo by Ben White on Unsplash

Summary

A new file format for SIMS and surface analysis is possible

- Some metadata terms still required
- Proprietary file formats are a barrier
  - Instrument vendor buy-in required
- Lack of awareness persists

Outlook

- Some low-hanging fruit
- Opportunity to make an impact
- Closed FAIR can still have benefits to industry



Please join the discussion at FAIRSpectra!

Thanks…



- For financial support
  - University of Manchester's Office for Open Research
  - SurfaceSpectra Ltd.

- For in-kind support
  - 101st IUVSTA Workshop (metadata workshop)
  - UK Surface Analysis Users Forum (UKSAF) (free exhibition space)
  - SIMS Europe (free exhibition space)
  - SpringSciX (free exhibition space)
  - NA-SIMS Workshop (metadata workshop)

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SIMS Europe

Spring SciX 2024

ZULIP